

Towards Artistic Image Aesthetics Assessment

CVPR 2023 Summary

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Motivation

- Existing aesthetics datasets (e.g., AVA) focus on **everyday photography**
- Artistic images are:
 - **More abstract**
 - **Diverse in style**
 - **Highly subjective**
- Traditional models struggle with artistic content

Key Contributions

- **Boldbrush Artistic Image Dataset (BAID)**: A large-scale dataset for artistic image aesthetics assessment (AIAA)
- **ArtisticsFormer**: A novel multi-task transformer model
- Joint learning of **aesthetic scores + style classification**
- State-of-the-art results on artistic image assessment

BAID: Boldbrush Artistic Image Dataset

- **60,337 artworks** from Boldbrush monthly contests
- **360,000+ votes** cast by online users
- **Score generation:**
 - Use raw vote counts per contest month
 - Normalize to **[0,10]** range (highest-voted → 10, lowest-voted → 0)

Vote-to-Score Normalization

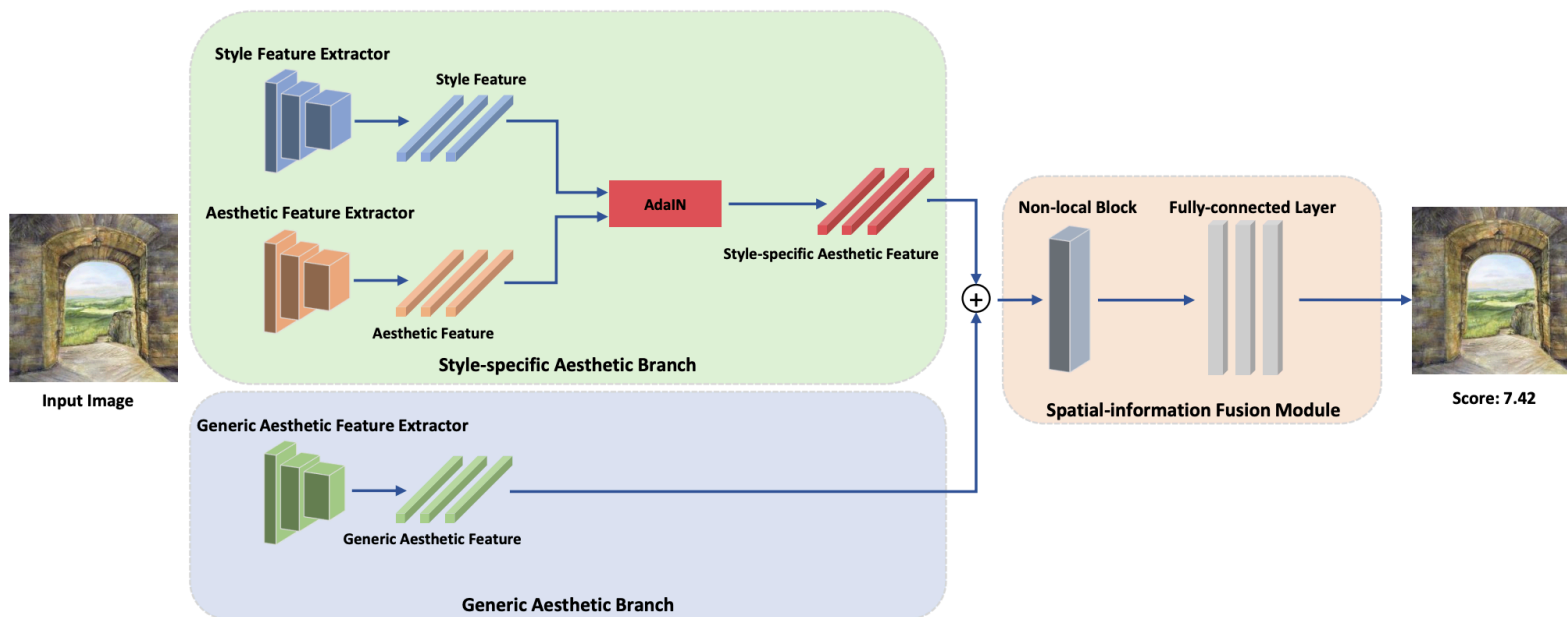
- Votes vary **greatly within each month** (range >200)
- BAID assumes contest artworks are **generally high quality**
- Normalization calibrates against **monthly context**

BAID Dataset Stats

Aspect	Value
Total images	60,337
Total votes	≈360,000
Aesthetic score range	0.0 - 10.0 (normalized)
Size	Largest AIAA dataset

SAAN: Style-specific Art Assessment Network

- Three-branch architecture:
 - Generic aesthetic branch** - extracts general quality features via self-supervised pretraining
 - Style-feature branch** - uses AdaIN to adapt to different artistic styles
 - Spatial fusion module** - integrates spatial (composition) cues via non-local block



Performance Summary

- SAAN outperforms conventional IAA models (e.g., NIMA, ResNet, ViT) on BAID
- Demonstrates effectiveness of combining **style-specific & generic aesthetic features**

Conclusion

- **BAID**: the first large-scale, vote-based aesthetic dataset for artwork
- **SAAN**: tailored model capturing both stylistic nuance and generic aesthetics
- Opens pathway for advanced artwork evaluation, curation, and style-aware generation

Thank You 🙏